(90) Inside Mach 33's SpaceX Valuation Model: Key Takeaways - YouTube

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Transcript:

(00:04) Welcome, welcome everybody. People are filing in here to Mach 33 private podcast. Welcome. All right. Uh we've got uh several folks joining in. Excited to uh get going. I don't know if anyone's been watching Twitter this morning. I am on the West Coast. I'm a little bit uh a little bit behind everybody.

(00:27) But um if you've been watching Twitter, you've been seeing all the interesting stuff with the road show officially kicking off. So, it's good time for us to be um discussing our initial models. I will say right off the kind of the bat um we have gone through many many many many iterations on this. Um it's something that when you're taking models and things change and new stuff comes in from the company obviously you have to take a lot into account.

(01:01) So um you know I take I take everything we're doing right now as think of it as you know final draft kind of versions and things are getting better over time. Um but we will probably be consistently iterating on this for the next week um going into IPO and things like that. So just um I'd like to lay that groundwork.

(01:18) I don't think we're phenomenally off on anything. I'm sure there's some tweaks that will happen, but just just kind of give you a sense of where we're at with our with our modeling. Um I don't know, Vlad, how many iterations? Hundreds at this point, right? Um >> yeah, I mean, it's it's been through four generations and that's that's getting to a point where I'm like, damn, I got to start again.

(01:43) >> So, we've done that uh quite a few times now. This is the fourth fourth time. >> Um and then each one, yeah, dozens each. So, yeah, there's been a lot of iterations. >> Yeah, it's um lots going on. So, I I just want to make sure we're all set up, you know, with that kind of context. Um we'll be releasing the um a writeup is the intention is to actually release the writeup after, you know, a little bit later today.

(02:10) Um and then the model will be available to, you know, the full model will be available to our customers. Uh I laugh when I say full model because it's you know of all these iterations it is quite large. So um it's a lot. So with that in with that in mind um we're going to go through kind of what we're seeing right now um where the numbers are kind of leading us more from like a you know put together Vlad's put together a lot of awesome HTML charts um to kind of like interact with some of the outputs of the model just so we can get a sense of where things are headed.

(02:40) Um, Vlad, did you want me to, you know, give you any any other kickoff or any other warm-up or did you want to just jump right straight into the charts? >> No, let's let's get stuck in. >> Um, >> awesome. So first of all, I'll say that this model um is is very different in one way to the last model in that in the last model, we tried to isolate what the key variables were and try to keep that list relatively short.

(03:17) So I'm pretty sure in the last one we had 20 30 variables that you could tweak. This one has 500 rows of variables. And so the the reasoning behind that is we decided to make it less user friendly but more customizable so that we can we can tweak things as um as SpaceX progress continues. So it's it's it's far more complicated. Um the the benefit is that we can be a lot more precise on a year-by-year basis and um it gives us more levers to move as as we discover more.

(03:56) So that's that's that's I'll start with that. Um I'm going to share some charts here. Hold on. So this is the the core cash allocation engine. So we split up SpaceX into three three business lines. Starlink, customer launch, and AI compute. Um, so you know, Starink, I'm pretty sure everyone here is pretty well accustomed to.

(04:28) Um, customer launch is a bit at this point is a bit of a blanket bucket for not only launches that are literally other companies, other space companies payloads, but also what could be pointto-point and Starfall and anything anything where you're not launching an internal uh SpaceX payload. And then AI compute is a mix of terrestrial data centers, orbital data centers, and the the applications that sit on top of those.

(05:04) So the way the engine works is um every year you have a starting amount of cash. um you you you net out the overhead the corporate level bills as well as we've because uh we've also carved out a bit of a bit for the moon and Mars and unfortunately it's it's hard well it's impossible to really calculate IRRa for the moon and Mars right now and so the methodology we use for the moon and Mars is accumulated book value like we did with the open source model that we had with ARC last year.

(05:44) Um, and because it's impossible to calculate uh an IRRa, you as a result can't rank by returns. And so that's a bit of an arbitrary carve out for the moon and Mars that that lives kind of on the corporate level that you pull out from last year's cash. Then after that, we rank each module by um how profitable investing capital into that module would be on a year-by-year basis.

(06:09) Based on that, we allocate cash to each module. So, we're here. So, we've allocated cash to each module. Um and then after that we we try to see if the modules can deploy the amount of capital that they've been allocated. And if they can't, which this this doesn't this chart doesn't really show, there's sort of a waterfall that redistributes the cache to the modules that can can allocate or you know invest that capital and then you build it and deploy it and the cycle starts again the next year.

(06:50) So that's kind of the the core engine of the the module that we've been building. Hey Vlad, before we uh continue, I wanted to remind folks because um >> I forgot to remind you guys all at the beginning. >> We have uh Q&A and obviously Vlad will be going through a lot of stuff. Um so feel free to drop Q&A in there.

(07:10) We may not be, you know, we'll try and answer as much as we can, I guess, is the point. We love Q&A. So there is the Q&A uh button, I believe, on your screens. It'll be at the bottom um tab there kind of thing. Feel free to drop questions in there as we're going and we'll try and we'll try and get them as we go. Yeah.

(07:30) So, this is what the revenue looks like. Um, in the base case, we haven't actually run the Monte Carlo yet. And part of the reason is that there's so many variables that we want to run the Monte Carlo in Python. Um, just because Excel won't be able to handle it. Uh so we're we're roughly here. This is S1 is living somewhere over here.

(07:58) Um so as you can see Starlink is the majority of revenue going up until 2030. And it says here V3 fully deployed. That's not quite the case. What what what that really is is uh Starlink reaching the demand curve saturation that we've modeled. So a point where launching more v3s isn't convincing enough in terms of IRRa and that's where you start deploying elsewhere.

(08:26) Um and so that's where you see that the AI compute site starts to take off. Um interestingly which I'll get to later you see that customer launch is this tiny slice sitting on top of everything else. But funnily enough the the actual IRRa of customer launch is really high. the the thing that stops uh customer launch ballooning is demand side.

(08:51) So that they're physically, you know, there's going to be thousands of Starship launches. And if that becomes the case, um there's a lot more volume that they can sell internally, sell uh to Starlink and AI compared to where we think the customer launch market is going. So, you know, take that with a grain of salt.

(09:15) We still have some work to do there to drill a little deeper. So we can see that um AI compute slice starts to grow and starts to match Starlink by 2035 and then after that starts to really take off. Now that's a mixture of the fact that by that point we expect the chips to become very very efficient and there'll be improvements on the the the satellite technology side.

(09:40) So what what we've done the way we've modeled the orbital compute satellites in this model is uh Elon set out the 100 kow per ton target that actually doesn't get hit until 2040 in the model. We start significantly lower. So around here the the satellite power density is roughly 50 kow per ton. Um, so as the satellites get more efficient, the chips get more efficient, this compute slice starts to really take off, we get to a whole, you know, companywide revenue 1.8 trillion in in 2040.

(10:19) Uh, one thing that I should mention here just to be honest is that um, the AI compute demand side is relatively unknown. I don't think there's anyone in the world who could really tell you what the demand for AI is going to be in in 20 240. Um, and when we built this model, we had to like really rein it in on the we had to almost build a semi arbitrary demand curve for AI just to to keep things sane.

(10:54) Um to give you a sense here when they reach uh 1.5 trillion in compute revenue that's I think it's somewhere between 300 and 400,000 compute satellites and you know they filed for 1 million. So that that gives you a sense um uh we still have to run the Monte Carlo like I said but when I shift the inputs of the model to the high end this goes to numbers that I won't even mention here because you'll you'll think we're crazy.

(11:23) So, um, I I had to rein this in a little bit, um, almost artificially, and that's because we have work to do probably over the summer to create some sort of demand curve for AI. Um, okay, Starink, drilling into Starlink. So, Starlink revenue is this axis and this purple line. You can see that around 2030 it kind of starts to plateau.

(11:51) This is because the V3 well it's it's because the demand curve that we've built the demand curve model that we built suggests that around then uh the return on launching more V3s is is going to get smaller. Um you can see the broadband so broadband is this slice the the darker green and DTOC is the lighter slice here. Um, broadband is the majority of Starink revenue up until 2013.

(12:20) Uh, and then direct to cell starts to to really kick in. So these are this is not revenue by the way. So these are subscribers. Let's let's just bear that in mind. The green slices are subscribers. Um we our thesis over the last year has been that direct to sell from from SpaceX's perspective the best strategy will be to once the V3 constellation is up um it it it's kind of like almost like the outside in strategy with broadband whereby they're going to try and make Starlink mobile your primary source of connection. So, wherever those

(12:58) DTOC satellites can um can serve you and give you speeds that that are more than suitable for what you would use when you're on the move, DTC is going to do that. And this is kind of the flip side of what most people believe about where the DTC market is going. Most people think that it's going to fill coverage gaps.

(13:22) And I think for a player like A Space Mobile that are partnering with with terrestrial Moss, that is the strategy. They're going to be there to fill coverage gaps where those M's can't fill. But we believe, you know, based on comments that Elon has made that in the short term that's going to be the case and it is the case today with T-Mo.

(13:45) So T uh the current generation of Starling Direct to Cell fills coverage gaps for T-Mo. But as that constellation rolls out, we think that they're going to try and move towards a business model where you on your phone when you you look at the top left rather than seeing AT&T and Verizon, you're going to see Starlink mobile.

(14:06) Um, and so we we've spoken about this in the past, but effectively that that that drives up um subscribers and it drives up ARPU because uh Starlink becomes your your main main source of connection where it can be. So if you're in the center of Manhattan, that's not going to happen. Well, it will happen, but the way it will happen is Starlink will wholesale from Moss um uh who have >> Yeah.

(14:36) Let me let me um I' I've had a lot of experience explaining this um to people asking me in the over the last few days. Um the you know if you do the model on the DTOC I think we've talked about this before but effectively you're somewhere between like you're getting somewhere between 10% and 1% 1 to 10% of the total customer value if you are a coverage gap filler.

(15:00) I'm oversimplifying here, but you're getting somewhere between one to 10% of the value. If you own the customer relationship, you can just backfill with um you get to flip the flip the equation where you get 90% to 99% of the customer value and then you just have to pay for the last 10 to one, you know, 1 to 10% with someone else. um it might be slightly different in the way they charge because maybe insurance you have a premium to that but you you know a premium on 1% is not a very big number even if it's a 3x right so um that's that's kind of the equation here

(15:33) and that's where customer ownership makes the most sense that's always been the SpaceX mentality and strategy I think currently we we we kind of all know that they're being they have partnerships and they need to build towards that but there there's that and then when you start to realize that this subs is like I think we've said you know effectively an opportunity for double counting.

(15:56) I think there's 400 and 400 million in each bucket but there's going to be a certain percentage of those maybe a pretty large percentage where it's um you know doing both broadband and um uh direct to cell. of this total customers may end up being something more like you know 500 600 600 million by uh 2040 but you have a larger chunk of share of wallet when it comes to connectivity of the entire connectivity share of wallet and you back fill in urban areas as needed for you know a certain percentage of your customers.

(16:27) >> Yeah. So um uh I'm a bit disappointed with how this chart came out. If I would have so I I've been working up and around the clock on this model and I wish I had more time to prepare these charts. If I would have done this chart and had enough time I would have made this line the subscribers and this these this stack bar the revenue but uh it came out the other way.

(16:50) And so uh what Aaron said is right is that these are not a lot of these um so this is staling subscribers this is TTOC subscribers we believe that a lot of these will be the same person in a way and so the actual total subscribers it depends if if we're talking by business line then this is correct there'll be one that it predicts 1 billion customers by 2040 across DTOC and Starlink but um it'll be it's uh yeah there's some cannibal cannibalization going on here in practice.

(17:27) So that's a good thing to mention. >> I'd also add something that's very intriguing interesting that I think most people don't realize is that um these customer counts kind of imply that that's where the bandwidth is going or there's that much bandwidth maybe an equal amount of bandwidth. The reality is the DTOC bandwidth is like almost um a rounding error compared to the uh the the broadband bandwidth from orbit.

(17:52) Um when you were to if you were to like forecast out to what this looks like in 2040, um it's it's really about that last mile connection. And so you don't need as much bandwidth. You know, I'm I'm not, you know, I don't want to get too much into the technical weeks, but it's an interesting implication here. And what you you spend so much time and effort getting a relatively small amount of bandwidth.

(18:13) I think it's less than 1%, isn't it, Vlad? In what we've modeled? >> Uh, for what exactly? >> The bandwidth for direct to sell >> of total bandwidth. >> Yeah. Of total bandwidth. >> Oh, yeah. Well, it's way less because it's different physics. >> Yeah, it's different physics. That's that's what I'm trying to highlight for people is it's um you get a lot less bandwidth DTOC.

(18:33) You could fill that up but your the vast vast vast majority maybe 99% of it is still broadband. Um so while what this really highlights if you think about those two things if you were to see the two charts side by side is it really highlights how important the DTOC is from a customer ownership perspective.

(18:53) Um and um you know what it can mean we believe here what it can mean for the um you know the opportunity for owning the customer relationship and then really driving up total users um versus um you know and revenues versus you know they just need a ton of bandwidth or something through the through through the actual phones. So anyway sorry I just want to bring that up.

(19:14) Another thing I want to mention before I move on is for people that may not be too well accustomed with what's going on here. Um, you might look at the 25 to 26 predictions and be like, "Okay, so nothing happens." And then doof, u, why is that the case? And I say this almost every week, but this is this is Starship and the next generation of satellites coming online.

(19:39) and they create uh almost an order of magnitude increase in capital efficiency uh when you combine both the improvements in the physics of the V3 satellites and the the capital efficiency of of Starship being a larger rocket that is fully reusable. And so we we'll have to watch. You know, the the S1 has clearly stated that they're going to start launching V3s in the second half of 26.

(20:06) In fact, this model I would say that this model is an oversimplification so far because 2026 I still have the model launching the V2s on Falcon and then 2027 all the cash goes to V3s on on Starship. And the reality is that it's probably going to be a little bit of a mix of both. And so maybe we'll see 26 a little higher, 27 a little lower, but the trend is clear once once it once things move towards Starship and the V3 satellites.

(20:39) It's it's a big inflection point. Um, so yeah. Okay. So wait, here we go. Here we go. So this is I'm going to explain everything here. The red line is the launch is here we go. Here Falcon launches a year. And you know, I can't really explain this dip. It's probably some sort of artifact of the cash allocation engine, but at this point, Falcon is launching only Starlink and then in 27, the model has Starship launching Starlink.

(21:12) So these bars are what is Starlink launching per year. Um, so Falcon 9 in my model goes from launching Starling to launching purely custom payloads. And we think that Falcon 9 is going to keep launching customer payloads into 2030 while Starship is not going to launch a single one. And the reason being, you know, there might be, you know, an odd one here or there, but um from a economics perspective, it makes much more sense for Starship to be launching internal payloads uh than launching customer payloads.

(21:50) So you see here, Starlink is the largest largest uh slice up until 2030 where if you think about the chart I showed you before in terms of revenue, we start hitting demand saturation for Starling V3. And at that point the Starlink launches go down and so well the company is still growing and so more Starships are being built.

(22:12) So where does that capacity go in 2031 2032? AI compute starts becoming a big slice and then this customer hold on this customer launch slice becomes um larger and larger as you go through here. And the reason is that um launching a customer payload on Starship requires a lot less capex than building AI satellites with uh the newest chips.

(22:49) No, because not only will the chips be expensive, but the compute satellites are in and of themselves going to be significantly more expensive than the Starling satellites. They require more efficient masstowe solar panels. They require next generation radiators that still need to be developed. And you know they're going to develop them.

(23:12) There are companies working on this exact technology now. Uh you know SpaceX are probably going to develop the same technology in house eventually. But launching a Starship full of those AI satellites is going to be much more expensive than launching a customer payload. And so we think that customer payloads are going to start, well, the model says that customer payloads are going to really start taking off in 2032.

(23:37) At that point, Falcon 9 is being wound down. Not a surprise because Starship is starting to bear that load. Um so yeah that's that's one thing and then you know over time we see AI compute starting to be a larger slice of the launches and I want to remind everyone that this customer launches slice is for now because we haven't we haven't dug deep enough here because it's not quite as loadbearing in terms of the financials.

(24:10) Customer launch may use up a lot of launches, but it it doesn't generate as much profit. It doesn't generate recurring revenue for one, but it doesn't generate as much profit as um uh launching Starlink or or AI compute satellites. Um so this bucket the the customer launch not only includes literally launching I don't know A satellites or Amazon LEO satellites whatever also includes uh conceptually speaking pointtooint uh and starfall so in space manufacturing also lives in this slice in our model >> and then uh you see the star >> I wanted to I think this is a good time

(24:54) you got a question here um you know asking why AI compute wouldn't it be a big customer for these launches you know other other AI computes um what do we think about that >> uh they would that would also that's actually a good point that would also live in the customer launch bucket because SpaceX aren't the ones that are developing and building the satellites which are which is what is very expensive and so there are there are two things by the way There are two things that probably don't come up in these slides but I'll mention is that we built

(25:28) a debt facility um module in this model where um it it it covers two two things. First of all, Terraab. So the debt facility says that SpaceX are going to need to to raise debt to build Terraab to a level where they can support the AI um aspirations. But then there's another debt facility line which is for the orbital data center satellites which even even though they're very expensive up front, they're almost just as profitable if not more profitable than Starlink in the in the long run. And so I just want to mention

(26:12) that that something that doesn't come up in these slides is that there's there's some debt facility lines that that that don't come up. And I think that launching other ODC's could very well fit in this customer launch slab in this stack bar. So that's a good question. I actually didn't even think of that. Um yeah, and so I want to explain why Starlink goes up again.

(26:39) So part of the reason is that all the V3s you've launched here are starting to de-orbit and at that point you need to launch more again to to keep keep us on the demand curve where we're at an optimal IRRa. That's one aspect. The second aspect is throughout this whole period of time the addressable market of Starlink is increasing.

(27:01) inflation goes up, GNI per capita goes up, the satellites get better and are and can serve a higher population density. So all those three things mean that your addressable market for Starlink in 2037 is significantly higher than it was in 2031. And so we're kind of building out the V3 constellation here. We're harvesting it in these years and redistributing launch capacity to uh XAI well SpaceX AAI orbital data centers and customer launch.

(27:38) And then over here we see Starink starting to um use up more more of the launch capacity again. Let me see if there's anything else I want to mention here. Hold on. Uh okay. I think maybe one thing to address is why are these launches going down here? Um I I think part of that reason is that throughout this whole time the Starship payload capacity is increasing.

(28:10) So over here in the model the payload capacity is 100 tons. By here it's 250 tons. So really this should have been up mass per year but it's done in launches per year. So that's part of the reason this is going down. Starship is getting better. And on top of that the payloads are getting more efficient.

(28:31) The satellites are more efficient etc. So that's I think that's why this is coming well I'm sure this is why this is coming down here. I think this is this one is just kind of like a rephrasing of a lot of the stuff that I covered before. But here you see the the share per module of Starship capacity. And it might be a bit easier to see it this way.

(28:54) So it starts Starlink and then when Starlink kind of matures, you see this big drop because then they're in the harvest period. The IR of launching more Starlinks drops. And throughout this time in our model, AI compute is still developing in terms of the technology of the satellites as well as the chips.

(29:13) And so that's why here you see that a lot of the excess capacity from stalling dropping doesn't actually go to AI compute uh it goes into customer. So this is when the customer opportunity opens according to the model 2031 to 2032. Uh and then up here is just the carve out which is a bit arbitrary but we assume that some of the launch capacity is going to go to the moon and Mars.

(29:36) Um, yeah. So, I think that this is a reiterating a lot of what I already covered. AI compute. Okay. It's a shame that this part is so small, but just so you know, terrestrial uh is a larger slice early on, which is the kind of light mauve slice here that sits in the middle. Um, the reason that terrestrial external revenue is so low is that the model carves out a large percentage of that for training.

(30:09) So, it actually builds quite a lot of terrestrial supply, but it says a lot of that's going to be training. And then we say that the the orbital data centers are going to be where the inference is sold. Um, so in a way to terrestrial data centers in our model are a bit of a a sync and so maybe that's why the the slice is is so thin when you look at it here and then the AI apps are the green slice on top.

(30:37) So that's you know literally everything sitting on top Grock um cursor or Grock's orchestration layer that will develop in place of cursor or both. Now the the Yeah. So that's that's that. And so I guess this shows that there's a lot of capacity to um either sell externally or gets absorbed by the AI apps. Um cuz you know Yeah.

(31:06) So I I'll I'll leave it at that. Here we've got free cash flow, Evidar and revenue all in in one chart. And then this line is the margin margin. >> Hey Vlad, before we before we jump too deep into uh this part, we had a good question on the AI SAT um bill of materials. Um we've referenced several times how it's expensive um you know uh maybe arguably more expensive.

(31:39) Um what are we expecting roughly for that bill of materials? like what what is >> like six million per satellite versus I believe somewhere between one and two million for Starlink. So in fact I can let me let me >> yeah I had the I was I was looking through our >> I was looking through our our model here in the background to validate those numbers. I knew it was fairly high.

(32:05) Um um but yeah uh the payloads are bit more expensive part of the reason why uh the terapab part of this project is so important um which I believe we'll touch on later right are we going to we've got later a conversation around terapab and cost in in these charts I can't remember >> potentially >> okay let me just address that question while you're doing that um someone's is asking a question I get a lot >> um which is what happens this is totally off off cue you know on a different conversation point but what happens if

(32:43) you know something happens to Musk what's the uh you know key man risk here um and I I've said this before in the past and you know it's an answer I don't think there is such a perfect answer I think um to this kind of question it's definitely a risk um Elon is the visionary Uh having said that, my most famous bad uh opinion on an investment was to sell Apple in like 2011 2012 time frame when they lost their visionary.

(33:17) Um because you can really underestimate what a someone who's trying to return value to the shareholder can do to the stock price. So I think that more than likely the consensus that I've gotten to and you know other smart people have gotten to is risk is huge. Mars probably gets hurt and then uh you know anything that requires a lot of vision and expense up front that people can't see the return in a short time frame probably gets hurt and then the things that are more return oriented probably get emphasized which is really good for you know

(33:47) shareholders. So I don't know probably a bad thing for humanity and a good thing for shareholders is is what that looks like at least in the short run. Uh short run being 5 to 10 years. Um all right glad I was uh answering one question. >> The the AI satellites in the first year that they launch we estimate will be around 5 million per satellite whilst the stalling satellites the V3s we assume will be roughly 1.2 million.

(34:14) So pretty big delta there. That being said, over time, >> it's that's about 50 kilowatts of of power um in the first iterations, right? >> 50 kow per ton. And so it would be 100 kilowatts at two ton. It almost doesn't matter. So we use a bit of an arbitrary mass because it doesn't really matter like you it what really matters is the kilowatts per ton.

(34:44) Um, and so what what what you'll find is the original ones will be I think in fact I'm pretty sure based on what we've heard um that the first ones will be roughly 50 kow per ton and then over time we're going to reach the 100 kows per ton. But at the same time, I think the costs are going to be coming down cuz SpaceX are going to be building their own supply chain uh developing these in-house more and more and and so that that's another compounding factor to think about really which is that the power density of the satellites will be increasing whilst the

(35:20) cost per ton of the satellite will probably be decreasing. Um but yeah I I think that the first ones will be 100 kow two tons roughly in the interest of time let's move to valuation Aaron so one thing that is funny it's not funny per se um but there's you know the market cap of the IPO is is being thrown around and you've got to really think about like what methodology should you be using at which point in time.

(36:01) Uh and there's there's there's reasons to use multiple methodologies. So you know we've we've calculated what our model outputs in terms of valuation in three methodologies. You got the revenue multiple, EVA multiple, and this kind of perpetuity growth uh methodology, Gordon's methodology, which is the perpetuity growth aspect of a DCF effectively.

(36:30) Um, and this chart here kind of shows what makes most sense at which point in time. So early on a revenue multiple probably makes the most sense um purely because everything everything that is EBID and below is pre-scale and as a result is diluting the potential growth of the company and so um you know revenue multiple probably makes most sense in the in the near term uh as as time goes on and uh the business is maturing, operating expenses become less detrimental, you know, you get more scale and so there's more flow through

(37:20) from the top line through to EBIDA and at that point EVIDA multiple is probably probably your best probably your best shout. um in the long run once a company matures then you probably want to start thinking when there's you know growth tapers off it's not going to grow much more you probably want to be thinking in terms of perpetuity growth so some sort of Gordon DCF methodology now if the um if the model had completely extended to where we think SpaceX is going to grow forever then this perpetuity growth methodology ology line

(38:00) would have probably come out on top. The reality is that our model goes until 20, it does go out to 2050, but we've only shown you 2040. And if you believe in the SpaceX thesis, it's probably going to go quite a bit further. And so, you know, I would I would suggest in my opinion that EV EBIT multiple is the the best way to think about this company between 2030 and 2040.

(38:25) Um and if you want to be conservative, use use perpetuity growth. But um yeah, sort of my two cents there. So this is what the valuations look like with each of those methodologies I just said. On the top you have ebida multiple um revenue multiple below and then perpetuity growth at the bottom. So in 2026 DCF says 2.

(38:55) 4 4 trillion today or end of 26. So not quite today. Um exit revenue sorry revenue multiple says 3.7 and the even ebid multiple says 5.6. 2030 ebidom multiple says 8.2. Exit multiple says 5.4. Revenue multiple says 5.4. And Gordon says 3.5. Let's just go straight to 2040 so I don't waste your time. In 2040, Ebot multiple says 18 trillion, revenue multiple says 10 trillion, and the DCF, which is probably the most conservative way to look at, says 6 trillion uh uh based on what we've modeled so far.

(39:42) And then, you know, as I said, I'm most I think the EBA multiple makes most sense for the next decade at least, if not two. And so, this is a sensitivity chart of um Oh, wait, no, sorry. This is this is Gordon. My bad. I wish I had ah did I not pull up the EBIT multiples? Well, let's go with Gordon.

(40:06) This is the most conservative valuations you can get uh based on our model. So today at 10% whack and an assumed perpetual by the way this goes through 2040. So if you think that in 2040 SpaceX is only going to grow 2.5% in perpetuity then the valuation today at a 10% whack is 2.47 trillion. If you think it's going to grow a little bit more then you go to 2.7 trillion.

(40:36) if you think that it's really risky and so the weighted average cost capital should be higher today you're at 1.75 and so that's how you should think about this this chart uh we use 10% whack in in our models and we actually use well that's the average we use different we use a different one for each re which business line uh but this is a simple way to look at it you know on the roll up of the whole And that's that's all I've got really Aaron.

(41:09) Um >> well we've got some questions here on terapab which I think is an important thing for us to cover. Um >> the someone's asking us if we accounted for the huge expense of terapab and uh the answer is yes. Um but I would like for you to kind of go a little bit deeper into how we did that. Um we did a analysis I believe it was last week that said uh kind of what we expect from a bottoms up um terapab cost to be.

(41:40) >> So um maybe we go a little bit into there and then also how we incorporated it into the model. >> Yeah. So, uh, one thing that's an interesting takeaway from our modeling is that the full terrafab doesn't have to be built until past 2040 to have enough chips for the revenue that I showed you on the compute side.

(42:07) That's that for me is a pretty big takeaway. >> It is a pretty big takeaway. Um so when he says what was it a million wafers >> uh the the 100,000 wafer starts per month >> the 100,000 wafer starts is onetenth of what we assume terrafab it was okay it was it was done in terms of compute it was like one one gawatt per year or something >> there's debate on what the one terowatt um >> the one ter yeah >> claim was and no one really quite knows it's really big.

(42:43) Lots of chips is what it means. So, everyone's kind of guesstimating what that what that functionally means in the future. But the bottoms up wafer starts per month I found to be quite helpful when we did that analysis because um 100,000 wafer stars per month. Like there's a whole bunch of different ways to do this. Like if you're doing like serverous style wafers, that's one that's like one chip.

(43:02) Um but like it's obviously uh you know could be bigger, could be smaller. kind of roughly estimated what the size of the chips are in any one of these wafers and all of that. And it came back to um that it's a lot of chips and it's going to be it's basically 100,000 wafer starts per month which again is supposed to be like the small version like almost not prototype but almost the prototyping level of this um would ended up ending up having enough chips for million optim optimum a year. I think it was uh like 2

(43:36) million cars a year and a whole buttload more for the the the compute um something like 40 million chips or something nuts. So like it clearly there's like a scale um thing here. We that did estimate a fairly high yield. So you know you could get first you know the very first yields being significantly less fairly high was like 70%.

(44:03) It's we're not estimating 99% or like TSMC level yields, but um you know that's that's that would be considered conservative at that level, but even if you got way way lower than that early on, you have more than enough chips. So it kind of gives a lot of leeway for chip yield to be quite low early in, you know, 2030 and still have more than enough chips for what's being trying to be done.

(44:25) But I think that's a pretty impressive takeaway there. Um interesting take anyway. So wait, there's there's a couple things that I want to mention here that I I was banging my head head against the wall when I was building this in the on the model side. So first of all, Terapab is a joint venture with Tesla. And so if they don't merge, then they will share the costs with Tesla because Tesla are going to consume a large portion of those chips one way or another, whether it's the cars or the robots.

(45:00) Um so that's something to bear in mind when you're modeling this that this all of this capex doesn't have to come out of SpaceX AI it's another story if Tesla is acquired obviously then but it's the same thing on net right so one thing to bear in mind is what portion of those chips are going to be used by Tesla versus SpaceX uh and as a result what portion of the terraab capex is going to go on which company's books So that's one one thing that I had to figure out.

(45:35) Um but yeah, the other thing is one terowatt worth of chips is a lot of chips. And I from my perspective, what the model says is we don't need anywhere near that many chips to hit the revenues that I showed you. We only need a tenth of that. And then in 2040, they kind of have to expand more.

(45:59) Um and so Terraab could be built to onetenth of what the aspirations seem to imply and orbital compute is covered through to 2040. That's my takeaway. I might be wrong. I might have some assumption that I've missed or miscalculated. But >> there's a question here talking about hey um this is all a lot of that compute is cap is relying on terapad to be online.

(46:25) about, you know, Radhard Nvidia chips and stuff like that. Um the the rising cost of all that I I wouldn't worry about about that. Yes, Terraab needs to be a thing. Um if you do any of the if you do any of the dives into this um semi analysis just came out with a um opal data center thing. It's I think it's funny and insightful that the guys were like the space guys.

(46:51) So we fixated more on the space side and thought that would be a bigger challenge. And there were the chip guys. They fixated on the chips. They thought that would be a bigger challenge. Um to me, you know, taking a step back, I think it's very obvious that if you um are you got to build the chips for the environment and they're they're just fundamentally different constraints.

(47:12) Um going hot and going um for power density is just not how these things are built today. So I think you have to kind of understand that that they're diverging priorities and they will be built differently. Um having said that the rad hardening and video trips like this is don't get descri like discouraged by rad hardening um conversations.

(47:36) People tend to fix it on that there are solutions in place already. There are um very easy to address solutions that are coming or already built. There's all sorts of solutions here that's like it's just a matter of how you want to design it. So >> one solution that I want to mention Aaron and let me know if you I don't know if me and you have spoken about this ever but um in sun-synchronous orbit the chips the sun is always kind of the same at the same angle to the chips.

(48:07) So you can just put a big slab of metal in between non radiation optimized space you know chips well just normal terrestrial chips put a big slab of metal in front of it and the thickness you know that's obviously creates more mass but um that's an easy solution in the short term I don't think there's a whole bunch hardening >> there there but like I I I don't want to leave that hanging because that'll look like we don't know what we're talking about there's secondary radiation effects um that come into effect like that come

(48:41) into play with something like that. But there point is there's a ton of solutions in place. You can do polymers, you can do just um redundancy on the chips level. Um like effectively solutions exist they will they as it's cost effective is kind of the early questions that everyone needs to understand.

(49:02) Um, I think it's a boogeyman that people that don't haven't like really, you know, paid attention to space for a while get all like to throw out there like it's a a scary thing. Um, it is a real problem. I'm not trying to say it's not a problem. Um, but it is it is a real problem. It has to work. All of these things correct.

(49:18) Um, but there's a lot of solutions in place and have been being built for a while to make that better and better over time. Um, and that's part of why they have that um I I for the cyclletron or something over in uh over on floor in the Florida coast where I normally sit. Um, literally to test these things like on the ground.

(49:38) It's one of the bigger cyclrons specifically for this and they're testing orbital compute. Um, uh, yeah, someone's asked me about the semi analysis AI compute. I read that yesterday, but I haven't really had time to truly digest it. I talked with one of the authors of it. I think what it comes down to is they're saying it's Oh, so the question is saying it's going to be 4x more expensive um to do compute in space versus on the ground.

(50:10) We were saying that that's going to it was more like two like they were saying right out of the gate. to 4x. I think we were saying something like I wanted to say at the current understanding it would have been maybe two and a half more expensive at today's rates and then we kind of estimated coming down uh in the 2030s. Their crossover said hey it could happen as the earliest is like mid 2030s.

(50:31) We're saying like very early 2030 kind of earliest it could happen. So clearly like a half decade off there. Um so >> yeah go ahead. Well, there's a couple things I want to say. I think this is a bit also the way that the question is phrased. It says 4x more expensive and won't be cheaper until 2040. Okay. Um but the there's there's two factors to think about that.

(50:56) So terrestrial is becoming more and more expensive but over time. In fact, Yensen Yensen said last week that he sees 1 gawatt of terrestrial compute costing 100 billion relatively soon. That's a lot. That's like 3x what it costs today. Um I don't know what the date was on that. Obviously it will eventually become that.

(51:21) But this is a guy who's like, you know, he's as close to it as it gets. And he's saying he's worried that terrestrials going to become really expensive. And then on the flip side, um, no one can reduce costs on space payloads as well as SpaceX can with their vertical integration, with their scale. And so this is kind of unprecedented on the space side.

(51:48) This is not like some startup designing a payload that is ingenious. this is a company that's going to scale these to like hundreds of thousands of satellites whilst bringing the you know bringing the R&D internally and developing it internally and so I think you know that you've got to think about the slopes of terrestrial becoming more expensive and orbital becoming cheaper no one knows the exact crossover um so that's one aspect second aspect is even if even if um it's more expensive.

(52:25) It may be impossible to scale on Earth to the same rates that you you can in space. In space, you have there's no real estate is free, energy is free. Um so there's that aspect as well. I can see a situation where it's more expensive, but they can just do a lot more than than I know that kind of goes against market dynamics >> to give semi- analysis credit.

(52:52) They've mentioned some of this stuff, right? Okay. Hey, they tried to caveat just like we did. I I think the core difference between the two analyses, and this is what I was referencing before, but I'll give more rough numbers because I need to digest a little bit deeper, but the core difference is they're they're saying like basically the IT the the chip stack going in, not just the chip, but all the IT stuff around it is 80 is really 80% of the cost of this thing.

(53:18) So, you really can't get away from So, what they're saying is really the power and all this other stuff, the land, you know, all that stuff is really just 20%. um we were saying it's more like uh 60%. So that's really the difference is like is it actually more on the getting plugged in and chips or is whatever. So like I we need to actually go and evaluate that.

(53:39) But that's why that's where you see a major difference, right? If if you're putting the same thing in. But this is also why it comes back to why the chip use case is so important because if you're building uh you know a rack to go into space, you're building it fundamentally for different reasons.

(53:54) And so there's a whole bunch of things like yes, if you're using today's chips and and all of that, it it ends up looking a little bit different, but fundamentally it's it comes down to if you're going to put them in space because your opex is cheaper than interesting to build it for that use case and actually optimize for that use case. It's all about density.

(54:13) It's a very different fundamental uh calculation and build out. Um, and for a group that's going to be very fixated on what today's supply chains look like, it makes total sense that you use you you'd probably do a much better job than we would understanding what the today's IT cluster looks like.

(54:30) With that said, the other thing is is that terrestrial is part of the SpaceX thing anyway. So, I'm not too worried about from a modeling perspective. They're already the cheapest on the ground, too. So, even if that crossover was a little bit later, they have the best terrestrial as well. I think that's the way it comes down to.

(54:46) Wait, I I have something pretty cool. So, I just apparently Goldwin is saying that space space space SpaceX AI's revenue is going to be 322 billion in 2030 and our figure is 305 billion. >> Goldman triangulation. >> Well, sell side's a little bit more bullish than uh you know, I think we're we're traditionally more buy side, I guess, but um >> sorry, I I just I got very excited reading that right now.

(55:13) Um >> um another thing >> hopefully as we optimize doesn't get worse >> is you know you said that uh they said that 80% of the cost is going to be chips that completely depends on whether you're buying your chips from Nvidia or getting the Mac cost from Terraab and to go back to the point before by 2030 terrafab will be producing enough chips for when uh OGC's are starting to take off.

(55:42) So terrafab the whole thing the one terowatt doesn't have to be complete they can be producing onetenth of the aspirational chip output and it will be more than enough >> for yeah where uh a ODC's are going to be in 2030. >> Yes. With that with that said um uh with that said looking at uh there's a there's a question here on on chips. Uh, I'm trying to triangulate here.

(56:10) I had a different thought and it's competing with the question I'm reading. So now my brain's like short circuiting. What I'll read this question. Why won't Nvidia or other novel chips that are being uh modified built right now allow competitors to get chips up earlier faster compared to SpaceX effort with terapab? Why not use these chips in the short run before terapab? The answer is well I'll give the quick answer Vlad.

(56:31) You can maybe get a considered answer. Um the the answer there would be yes they will. Um I would be very surprised if there are TPU sort of conversations happening. Um you know with Google uh TPU tends to favor it. It's a fairly high level. I'm not like in the TPU thing is super deep but it tends to favor the power density.

(56:56) So that would be great for that would be great for using those on orbit. Um AI5 and working with Intel, Samsung and all these things I think are probably going to be the way that they get to early iterations on orbit. Um Nvidia is doing the Space One chip. I forget is that the name of it? Bear Rubin Space One or something.

(57:16) >> Yeah. Um so they're doing that as well. The answer is yes. They can do do any and all the above. early on it's going to be more about how do you get compute to power at ASAP. If it's harder to get compute whatever is on whatever is available compute to power on the ground um as these things become more space optimized compute to power um may kind of flip they you may not even care how much more expensive it is to get it up initially because you can get it up faster.

(57:47) So you know compute to power timeline is is in theory quite important. I think if we back up one second, everything goes to the lens of the demand curve on on AI compute or just AI in general. And Vlad kind of referenced that earlier, a demand curve that's as far as like from our perspective, infinite >> changes a lot of things and and and drives a lot of decision-m.

(58:11) So, we have to like keep that in mind. >> Well, we're up on time. So, let me leave let me leave you with this other thing. Uh, whoever asked that question, there's one more aspect. Um the technology for orbital data center satellites is not it's in like prototype level but it's not where it needs to be to be at cost parity anyway.

(58:33) Cowboy Space and StarCloud they can launch all the payloads they want with current Nvidia chips but it economically won't be as profitable as when it really takes off which is in 2030. And so there's a lot of R&D that needs to be done in terms of finalizing designs in the next couple years, but more importantly developing the supply chains to be able to build these at reasonable volume.

(59:00) And by that time, Terraab is ready. And so there's that aspect to think about as well. Um, on that note, uh, we've gotten confirmation that one of our one of the people we know who was the lead engineer on the thermal side of Starlink for V1 and V1.5, we've been asking him about the thermal side, not just the radiators, but the whole thermal side of these ODC satellites.

(59:27) he's agreed to come on to the podcast and so either we'll have him on next week or in the coming weeks and we can roast him on all of the technology side of these ODC satellites because he he'll know better than anyone else. So there you go specifically from the perspect premature right now. That's what I'm saying. >> Yeah, it's uh there's there's plenty of of things to do.

(59:50) We have one question here. I I know we're over but like one last question. How much higher is your current valuation estimate for the EBITDA multiple or model? EBITDA model. Um >> as opposed to >> that's all we got. So the question maybe as opposed to the per perpetuity growth. Um >> oh no much higher versus what the IPO is I guess.

(1:00:14) >> Yeah. >> Um okay. Oh versus perpu. Wait I'll show you right here. Hold on. So, Perpetuity says it's 2.5 trillion, which is roughly in line with where this IPO is probably going to land roughly. Uh, the E the Evod multiple says 5.6. Take it with a pinch of salt. >> Yeah, it's a pinch definitely a pinch of salt.

(1:00:41) I think um you know, anything with SpaceX, you know, despite what all of the um the skeptics say, you have to you're looking at buying, you know, future cash flows. Um, I don't know how you do a DCF on historic cash flows. Well, is that does that work that way? >> I mean, >> that's a joke. >> Um, that's I'm saying historic cash flows. You got discounted back, I guess.

(1:01:03) But, um, you know, anyway, point being, uh, >> yeah, you got to be thinking about the future here. Obviously, our our estimates are a little bit more, you know, hey, you see the big elbow in our estimates based on bandwidth coming on online. >> Um, And also that's what yeah 2040 IBIDA bear that in mind. >> Yeah 2040.

(1:01:26) So yes we're looking further we're looking further into the future. >> Um you know there's definitely it's hard for people to imagine anything big here because you know there this is already in the trillions and the biggest company now is in like the four to five trillion range. It's so okay it just feels arbitrary that that is the anything beyond that is is even possible.

(1:01:47) So I with that being said, I think it's it's important for people to to bear that in mind. Um you are trying to figure out what the what the future looks like. We're trying to be less wrong than everyone else is kind of the goal here. Um and I think based on the the the physics and engineering and and what we believe to be, you know, happening over the next 5 to 10 years, it this is reason.

(1:02:11) All right, Vlad, any you want you want the final word? No. All right, cool. Thanks everybody for joining. I appreciate all the questions. It's been incredibly helpful and interesting. Uh we will be updating this probably all the way through the road show, which is really just a week. So, it'll be it'll be a rough kind of time.

(1:02:28) We're expecting to put an initial kind of write up um later today, but um hopefully um you know, hopefully we've got uh you know, helpful information out here for you and you know, feel free to give us feedback as as needed. Thanks everybody.